

AI-Assisted MRI Classification for Insurance Decision Support

Individual Paper for STATS 421

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Purpose of Report: Client meeting with an insurance company on medical-image review, claims
triage, and model risk management

Audience: Insurance company professionals with limited statistical background

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Introduction

While insurance companies do not diagnose patients, they regularly interpret medical evidence when reviewing claims, prior authorizations, care management cases, and payment decisions. Medical imaging is often one of the most important forms of evidence in these workflows because it can influence the urgency of a case, the expected treatment path, and the amount of human review that is needed. At the same time, MRI records can be difficult to evaluate quickly, especially when many claims arrive from different providers and when documentation is incomplete or inconsistent. For this reason, it may be of interest whether artificial intelligence can support a more efficient and consistent review process while still keeping final medical judgment in the hands of qualified clinicians.

To help the insurance company evaluate whether this approach can support its review process, this report examines the use of brain MRI images to classify four diagnostic categories: glioma, meningioma, pituitary tumor, and no tumor. The analysis compares a simple convolutional neural network with a more advanced transfer learning model based on EfficientNetB0. The statistical question is not only whether the model can produce a high overall accuracy, but also to answer the more practical question of whether the model can identify difficult tumor categories more reliably, reduce unnecessary manual review, and flag cases that deserve closer attention.

We explore this question through a public brain tumor MRI dataset which gathered 7,021 total MRI images, with 4,570 images used for training, 1,140 for validation, and 1,311 for testing. Every image is resized to 224 by 224 pixels with three color channels so that the images can be processed in a consistent format. The four outcome classes are treated as a multi-class classification problem.

From an insurance focus, this research should be understood as decision support rather than automated decision making. A model of this kind could help prioritize complex claims, support a second review of submitted medical records, and help identify cases where additional documentation may be needed. Therefore, the central purpose of this report is to explain what the model does, how it was evaluated, what the results mean for an insurance workflow, and what safeguards would be required before practical use.

Statistical Background & Terminology

With this area of medical research, the primary statistical goal is what we describe as classification. In a classification problem, the goal is to assign each observation to a category as correctly as possible given the data. Here, each observation is an MRI image, and the possible categories are glioma, meningioma, pituitary tumor, and no tumor. Since there are more than two possible outcomes, this is a multi-class classification problem that estimates which of several labels is most likely for the image. In an insurance setting, this is similar to sorting incoming cases into meaningful review groups so that the company can decide which cases require immediate medical review and which cases are more routine.

In order to create an effective classification model, we must run this model through training, validation, and testing. In the first stage, training data are used to teach the model. From there, we move on to the next stage with our validation data to check whether the model is improving without simply memorizing the training examples. Finally, the test data are held out until the end so that the final performance can be evaluated on images that were not used to fit the model. This separation is important because those interested in medical records data do not need a model that only performs well on old examples and instead prefer a model that can

generalize to future claims and medical records from providers that may not look exactly like the original dataset.

The baseline model in this project is a convolutional neural network, often called a CNN. A CNN is a type of deep learning model that is designed for images. Rather than asking the analyst to manually measure every possible visual feature, the model learns patterns from the pixels themselves. In early layers, the model may learn simple patterns such as edges, curves, and changes in brightness. In later layers, it can combine those simple patterns into more complicated features that help separate one tumor type from another. This is useful for MRI images because the useful information is visual and spatial, not only numerical.

The advanced model uses transfer learning. Transfer learning means that a model begins with knowledge learned from a very large image dataset and is then adjusted for a smaller, more specific task. There are many advanced transfer learning models out there, but one model of interest is the EfficientNetB0 model.. Expanding beyond brain MRI scans, the model brings useful low-level visual knowledge, such as detecting boundaries, texture, and contrast. Instead of starting from a blank page, it applies general visual experience and then adapts that experience to classification tasks.

Several measures are used to evaluate the model. Accuracy shows the proportion of MRI images classified correctly overall. However, accuracy alone may hide errors within specific tumor categories. Therefore, precision is also used to show how often the model is correct when it predicts a particular class. Recall asks, among the true cases of a class, how many the model successfully finds. A hybrid metric called the F1-score combines precision and recall into one balanced measure. Finally, the ROC-AUC, short for Receiver Operating Characteristic–Area Under the Curve, measures how well the model separates one class from the other classes across

different decision thresholds. From both a medical and insurance viewpoint, these measures matter because different errors have different business and ethical costs. A false negative, such as failing to flag a true tumor case, can delay review of a serious claim. A false positive can send too many routine cases to manual review and reduce operational efficiency.

One major concern from training models based on these measures is the precaution of overfitting, which is when a model learns details that are too specific to the training data and then performs worse on new data. A model with high training accuracy but weak test performance is not trustworthy for real-world use. To reduce this problem, the project uses numerous machine learn techniques (validation data, dropout, batch normalization, data augmentation, and early stopping, all fully detailed in the appendix), all which share a goal to teach the model stable patterns that are likely to appear in future MRI images as opposed to random details from the original dataset.

Methodology

The methodology begins with the structure of the data. Each MRI image is resized to 224 by 224 pixels and normalized so that pixel values fall within a common range. This makes training more stable because every image is presented to the model in the same numerical format. Training images are slightly transformed (data augmentation) through changes such as rotation, shifting, zooming, brightness adjustment, and horizontal flipping, which do not change the medical category of the image, but instead create more visual variation. As a result, the model is less likely to rely on one exact image position or lighting pattern.

The project materials describe the four tumor categories as balanced in the prepared data, with each class making up about one quarter of the sample. In this study, the balance across glioma, meningioma, pituitary tumor, and no tumor make the accuracy measure easier to

interpret. It also makes class-specific measures such as recall and F1-score more meaningful, because the model has comparable exposure to each category.

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Class Distribution Percentage:
glioma: 25.00%
meningioma: 25.00%
notumor: 25.00%
pituitary: 25.00%
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Figure 1. Reported class distribution in the prepared MRI image data.

The first candidate model is a baseline CNN. The baseline contains a sequence of convolutional layers, ReLU activation functions, max-pooling layers, a flattening step, a dense layer, and a final softmax layer for the four classes. The model is trained with the Adam optimizer and categorical cross-entropy loss. This model is a reasonable starting point because it is designed for image recognition and can learn visual features directly from MRI images. However, it also has a limitation: it must learn both basic image features and the final classification rule from this dataset alone.

The second candidate model is an EfficientNetB0 transfer learning model. The final classification layer from the pretrained model is replaced with a new structure designed for the four MRI classes. The updated model includes global average pooling, batch normalization, dropout, a dense layer, and a softmax output layer. The training process is performed in two phases. In the first phase, the pretrained backbone is frozen, so only the new classification layers are trained. In the second phase, parts of the pretrained network are unfrozen and fine-tuned with a smaller learning rate. This two-stage process is useful because it first preserves the general visual knowledge in the pretrained model and then carefully adapts it to the MRI task.

The working hypothesis is that transfer learning will outperform the baseline CNN, especially for tumor types that are visually harder to distinguish. This hypothesis is reasonable

because the baseline CNN has to learn everything from the available MRI data, while the EfficientNetB0 model begins with a stronger visual feature system.

The evaluation strategy uses several complementary metrics instead of one final score. Accuracy is used as the broad summary of correct classification. Precision, recall, and F1-score are used to understand the model's behavior for each tumor class. ROC-AUC is used to evaluate class separation across thresholds. In a stricter deployment study, an insurance company should also use confidence intervals, external validation, and a paired comparison test such as McNemar's test to determine whether the difference between models is statistically reliable on a new claims population. The current project should therefore be read as a model comparison and proof of concept, not as a complete clinical or actuarial validation.

Several assumptions must be considered. The labels in the dataset must be correct enough to train a reliable model. The training, validation, and test sets must be separated properly so that the same patient or nearly identical images do not appear in both training and testing. The images should also be representative of the type of MRI evidence that the insurance company would receive from providers. If these assumptions are not satisfied, the reported accuracy could be too optimistic. For this reason, a future insurance implementation would need an additional validation study using the company's own historical claims records, with privacy protection and clinical oversight.

Results

The results show a clear improvement when moving from the baseline CNN to the EfficientNetB0 transfer learning model. The baseline model reaches an overall accuracy of about 70 percent. By comparison, the advanced model reaches an overall accuracy of about 85 percent. This difference is important because it suggests that transfer learning provides useful additional

structure beyond the simple CNN. However, the more important result is not only the increase in total accuracy. For an insurance company, the class-specific results are more informative because they show where the model is strong and where it still needs human supervision.

BASELINE CLASSIFICATION REPORT				
	precision	recall	f1-score	support
glioma	0.67	0.52	0.58	400
meningioma	0.60	0.40	0.48	400
notumor	0.72	0.98	0.83	400
pituitary	0.74	0.89	0.81	400
accuracy			0.70	1600
macro avg	0.68	0.70	0.68	1600
weighted avg	0.68	0.70	0.68	1600

Figure 2. Baseline CNN classification report.

	precision	recall	f1-score	support
glioma	0.87	0.75	0.81	400
meningioma	0.80	0.73	0.77	400
notumor	0.88	0.97	0.93	400
pituitary	0.84	0.94	0.88	400
accuracy			0.85	1600
macro avg	0.85	0.85	0.85	1600
weighted avg	0.85	0.85	0.85	1600

Figure 3. EfficientNetB0 transfer learning classification report.

Overall, EfficientNet outperforms the baseline CNN across the main evaluation measures, improving overall accuracy from about 70% to about 85%. The improvement is especially important for glioma and meningioma, where recall rises from about 0.52 to 0.75 and from about 0.40 to 0.73, making the model more useful as an insurance review-support tool.

The F1-scores show the same pattern. For the baseline model, glioma has an F1-score of about 0.58 and meningioma has an F1-score of about 0.48. Under the advanced model, these increase to about 0.81 and 0.77. This matters because F1-score rewards a model that balances finding true cases with avoiding too many incorrect positive predictions. In practical insurance language, the advanced model is better at finding cases that should be reviewed while also maintaining a more reasonable level of accuracy among the cases it flags.

The ROC-AUC results also favor the transfer learning model. For the baseline CNN, glioma has an AUC of about 0.82 and meningioma has an AUC of about 0.79. For the advanced model, these values rise to about 0.93 and 0.95. The no-tumor and pituitary classes are strong in both models, but the advanced model still produces very high AUC values for these categories. The ROC curves suggest that the transfer learning model separates the tumor classes from the

other categories more effectively, which could help an insurance company choose thresholds for different levels of manual review.

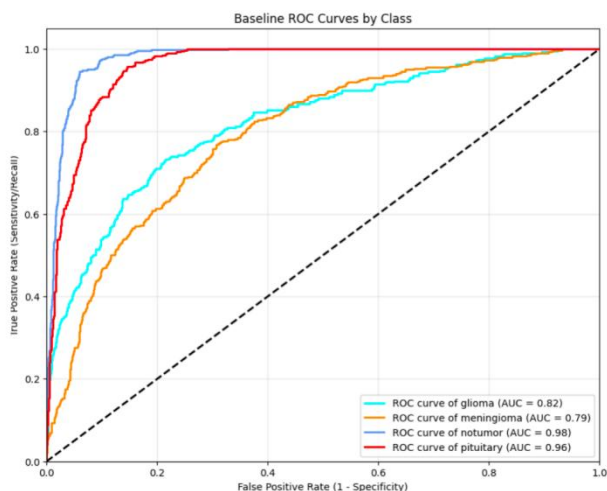


Figure 4. Baseline ROC curves by class.

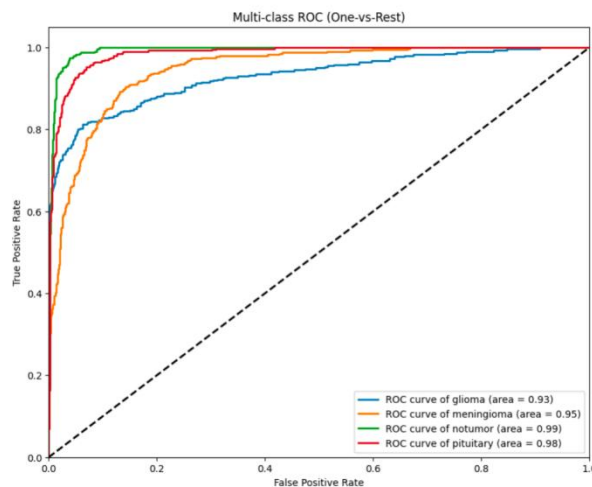


Figure 5. Advanced model ROC curves by class.

The training and validation curves further support the conclusion that the advanced model generalizes better than the baseline model. The baseline model improves over training, but its performance remains lower and its difficult-class recalls are weak. The advanced model shows a smoother improvement pattern and reaches a stronger validation performance. This suggests that the pretrained visual features and fine-tuning process help the model learn patterns that are more stable across images.

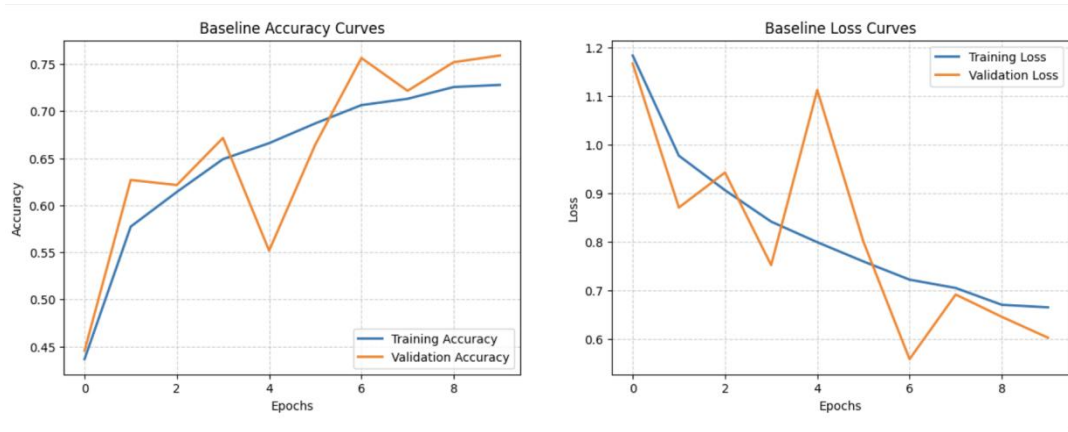


Figure 6. Baseline model accuracy and loss curves.

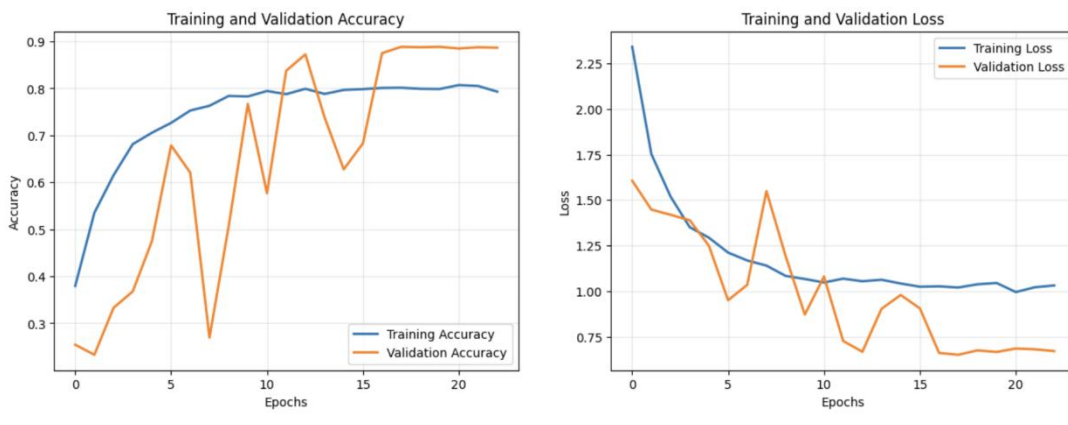


Figure 7. Advanced model accuracy and loss curves.

From an insurance perspective, these results support a limited but valuable recommendation. The advanced model could be used as a decision-support tool for case triage, quality assurance, or second review. For example, cases that the model identifies as likely tumor cases could be routed more quickly to a medical reviewer. Cases with low model confidence could be flagged for additional documentation. However, the model should not be used to make coverage decisions without human review. The remaining errors, especially between visually similar tumor types, show that the model is not reliable enough to replace a radiologist's report or a clinical policy review.

Conclusion

This paper reframes a brain tumor MRI classification project for an insurance company audience. The project compares a baseline CNN with an EfficientNetB0 transfer learning model on a four-class MRI image classification task. The main finding is that the transfer learning model performs better across the most important measures. Overall accuracy increases from about 70 percent to about 85 percent, and the biggest practical improvements occur in the more difficult glioma and meningioma classes. These improvements are important because difficult tumor classifications are precisely the cases where an insurance company would want stronger review support and fewer missed signals.

The results also show why accuracy should not be the only measure used in a medical-image workflow. The baseline model has acceptable overall performance, but it misses many glioma and meningioma cases. The advanced model improves recall, F1-score, and ROC-AUC for these classes, which makes it more suitable for a triage role. In practical terms, this model could help an insurance company prioritize cases, reduce delays in manual review, and identify claims that may need additional medical documentation. These benefits are operational and administrative, not a replacement for medical judgment.

At the same time, the correct conclusion is cautious. The advanced model is better, but it is not perfect. Some MRI images remain difficult because tumor types can look similar, image quality can vary, and the public dataset may not fully match the population of records received by an insurance company. Before any real use, the company should validate the model on its own historical claims data, confirm that the training and testing data do not contain leakage, monitor performance across providers and patient groups, and require clinician review for any

case that affects patient care or coverage. Therefore, the recommended use is not automated claim approval or denial. The recommended use is carefully governed decision support.

Overall, transfer learning provides a meaningful improvement for brain tumor MRI classification and is the better modeling approach in this project. For an insurance company, the model's strongest value is its ability to support faster and more consistent review of complex medical imaging evidence. The model should be treated as a tool that helps professional reviewers focus their attention, not as an independent authority. With external validation, threshold calibration, privacy safeguards, and ongoing monitoring, this type of model could become a useful part of a responsible insurance analytics workflow.

Appendix

Appendix A: Dataset and Model Summary

Dataset source: Public Brain Tumor MRI Dataset from Kaggle.

Reported total images: 7,021 MRI images.

Reported split: 4,570 training images, 1,140 validation images, and 1,311 test images.

Input dimensions: 224 x 224 x 3 after resizing.

Outcome classes: glioma, meningioma, pituitary tumor, and no tumor.

Client interpretation: The dataset is appropriate for a proof-of-concept model comparison, but it is not enough by itself for an insurance deployment decision.

Appendix B: Algorithm Outline

Step 1: Load MRI images and labels from the prepared dataset folders.

Step 2: Resize images to 224 x 224 x 3 and normalize pixel values.

Step 3: Apply data augmentation to the training images, including rotation, shifts, zoom, brightness adjustment, and horizontal flipping.

Step 4: Train a baseline CNN using convolutional layers, ReLU activation, max pooling, dropout or batch normalization, and a softmax output layer.

Step 5: Train an EfficientNetB0 transfer learning model by replacing the final classification layer and training the new head while the backbone is frozen.

Step 6: Fine-tune the advanced model by unfreezing selected pretrained layers and using a smaller learning rate.

Step 7: Evaluate both models using accuracy, precision, recall, F1-score, ROC-AUC, and training and validation curves.

Step 8: Interpret the results for insurance decision support, with special attention to false negatives, difficult classes, and human review requirements.

Appendix C: Main Numerical Results

Measure	Baseline CNN	EfficientNetB0 Transfer Learning	Insurance Interpretation
Overall accuracy	About 0.70	About 0.85	The advanced model gives a stronger general triage signal.
Glioma recall	About 0.52	About 0.75	The advanced model misses fewer true glioma cases.

Meningioma recall	About 0.40	About 0.73	The advanced model greatly reduces missed meningioma cases.
No-tumor recall	About 0.98	About 0.97	Both models identify this class well.
Pituitary recall	About 0.89	About 0.94	The advanced model improves an already strong class.
Glioma AUC	About 0.82	About 0.93	The advanced model separates this class more clearly.
Meningioma AUC	About 0.79	About 0.95	The advanced model gives a much stronger ranking signal.

Note: The project slides report a test set of 1,311 images, while the classification report screenshots display support values totaling 1,600. Before final submission or any deployment discussion, this count should be checked against the original notebook output. The performance pattern is still clear: the advanced model improves the difficult glioma and meningioma categories.

Appendix D: Glossary of Technical Terms

Accuracy: The proportion of all observations that the model classifies correctly.

Batch normalization: A training technique that stabilizes the numerical scale of activations inside a neural network.

Categorical cross-entropy: A loss function used when a model predicts probabilities across multiple classes.

Classification: A statistical or machine learning task in which each observation is assigned to one of several categories.

Convolutional neural network: A neural network architecture designed for image data, using filters to learn local visual patterns.

Data augmentation: The process of creating modified versions of training images to improve generalization.

Dropout: A regularization method that randomly removes some neural network units during training to reduce overfitting.

EfficientNetB0: A pretrained image classification architecture that balances predictive strength and model size.

F1-score: The harmonic mean of precision and recall, used when both false positives and false negatives matter.

Fine-tuning: The process of updating a pretrained model on a new, more specific dataset.

Freeze: To keep existing model parameters fixed during training.

Overfitting: A situation in which a model performs well on training data but poorly on new data.

Precision: Among cases predicted to belong to a class, the proportion that truly belong to that class.

Recall: Among true cases of a class, the proportion that the model correctly identifies.

ROC-AUC: A measure of how well a model separates one class from other classes across thresholds.

Softmax: A function that converts model outputs into predicted probabilities across multiple classes.

Transfer learning: The practice of adapting a model trained on one large task to a new related task.

Unfreeze: To allow previously fixed model parameters to be updated during training.

Appendix E: Works Cited

Masoud Nickparvar. Brain Tumor MRI Dataset. Kaggle.

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

Project codebase: [https://github.com/georgewuu/stats_426-Final-](https://github.com/georgewuu/stats_426-Final-Project/blob/7de296fec7526d89b03b2dd8c6255bc300fb5331/stats426_project.ipynb)

[Project/blob/7de296fec7526d89b03b2dd8c6255bc300fb5331/stats426_project.ipynb](https://github.com/georgewuu/stats_426-Final-Project/blob/7de296fec7526d89b03b2dd8c6255bc300fb5331/stats426_project.ipynb)